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Chapter 14

Bond Prices and Yields

INVESTMENTS

THIRTEENTH EDITION

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Chapter Overview

- **Debt securities** (*fixed-income*) promise either a fixed stream of income or one determined by a specified formula.
- **Bond markets**
 - Treasury, corporate and international bonds.
 - Bond pricing and returns.
- Impact of default or credit risk on bond pricing.
 - Fixed-income derivatives.
 - Credit default swaps.
 - Collateralized debt obligations.

Bond Characteristics

- **Bond**
 - Security that is issued with a legally binding borrowing arrangement.
 - Issuer agrees to make specified payments to the bondholder on specified dates.
- **Par value (Face value)** is the payment to the bondholder on the bond's maturity date.
- **Coupon rate** is a bond's interest payments per dollar of par value.
- **Bond indenture** is the contract between the issuer and the bondholder.

Treasury Bonds and Notes

- Maturity
 - Treasury notes—1–10 years.
 - Treasury bonds—10–30 years.
- Both bonds and notes may be purchased directly from the Treasury.
- Denominations.
 - As small as \$100, but \$1,000 is more common.

Table 14.1 Prices and Yields of U.S. Treasury Bonds, August 15, 2021

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U.S. Treasury Quotes						
MATURITY	COUPON	BID	ASK	CHANGE	ASKED	YIELD (%)
Aug 15 24	2.375	105.8188	105.8313	0.030		0.417
Aug 15 26	1.500	103.4563	103.4750	0.064		0.790
Aug 15 30	0.625	94.6938	94.7250	0.186		1.246
Aug 15 40	1.125	88.4125	88.4750	1.002		1.847
Aug 15 40	3.875	134.1875	134.2500	1.748		1.749
Nov 15 44	3.000	120.4500	120.5125	1.830		1.904
Feb 15 47	3.000	121.9750	122.0375	1.280		1.905
Aug 15 49	2.250	106.6625	106.7250	1.948		1.938

Corporate Bonds

- **Callable bonds** typically come with a period of call protection.
- **Convertible bonds** give holders option to exchange each bond for a specified number of shares of the firm's stock.
- **Put bond** gives holder option to exchange for par value at some date or to extend for a given number of years.
- **Floating-rate bond** has interest rate that is reset periodically according to a specified market rate.

Table 14.2 Listing of Corporate Bonds

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ISSUER NAME	SYMBOL	COUPON	MATURITY	MOODY'S®/S&P		HIGH	LOW	LAST	CHANGE	YIELD %
AMAZON COM INC	AMZN5182954	0.450%	May 12 24	/AA		99.913	99.739	99.913	-0.176	0.482282
HORMEL FOODS CORP	HRL5193038	0.650%	June 03 24	/A		100.193	99.622	100.122	-0.061	0.491393
GENERAL MTRS FINL CO INC	GM5198643	1.500%	June 10 26	/BBB		100.822	99.811	99.903	-0.872	1.520917
BP CAP MKTS AMER INC	BP4763976	3.790%	Feb 06 24	A1/		107.388	107.34	107.34	-0.116	0.655786
GLAXOSMITHKLINE CAP INC	GSK4632356	3.375%	May 15 23	A2/		105.279	105.214	105.228	-0.028	0.328114
INTEL CORP	INTC4969549	4.750%	Mar 25 50	A1/A+		134.693	134.275	134.596	0.365	2.933182

Figure 4.2 Listing of corporate bonds.

Source: FINRA (Financial Industry Regulatory Authority), August 24, 2021

Preferred Stock

- Considered to be equity, but often included in the fixed-income universe.
- Like bonds
 - Promises to pay a specified cash flow stream.
 - Commonly pays a fixed dividend.
 - Rarely gives holders full voting privileges in firm.
- Unlike bonds
 - Failure to pay the promised dividend does not result in corporate bankruptcy.
 - Dividends owed cumulate.

International Bonds

Foreign bonds

- Issued by a borrower from a country other than the one in which the bond is sold.
- Denominated in the currency of the country in which it is marketed.
 - Samurai bonds.
 - Bulldog bonds.

Eurobonds

- Denominated in one currency, usually that of the issuer, but sold in other national markets.
 - Euroyen bonds.
 - Eurosterling bonds.
- Not regulated by U.S. federal agencies.

Innovation in the Bond Market

- **Inverse floaters** are like floating-rate bonds, except coupon rate *falls* when the general level of interest rates *rises*.
- **Asset-backed bonds** use income from a specified group of assets to service the debt.
- **Catastrophe bonds**' final payment depends on whether there has been a specified catastrophe.
- **Indexed bonds** make payments that are tied to a general price index or the price of a commodity.
 - Treasury Inflation Protected Securities (**TIPS**).

Table 14.1 Principal and Interest Payments for TIPS

- e.g. coupon rate TIPS with coupon rate of 4%

Time	Inflation in Year Just Ended	Par Value	Coupon Payment	+	Principal Repayment	=	Total Payment
0		\$1,000.00					
1	2%	1,020.00	\$40.80		\$ 0		\$ 40.80
2	3	1,050.60	42.02		0		42.02
3	1	1,061.11	42.44		1,061.11		1,103.55

First Year Nominal Rate of Return:

$$\text{Nominal return} = \frac{\text{Interest} + \text{Price appreciation}}{\text{Initial Price}} = \frac{\$40.80 + 20.00}{\$1,000} = 6.08\%$$

First Year Real Rate of Return:

$$\text{Real return} = \frac{1 + \text{Nominal Return}}{1 + \text{Inflation}} = \frac{1.0608}{1.0200} = 4.00\%$$

Bond Pricing ¹

$$\text{Bond Value} = \sum_{t=1}^T \frac{\text{Coupon}}{(1+r)^t} + \frac{\text{Par Value}}{(1+r)^T}$$

- First term of right-hand side of equation is the present value of an annuity.
- Second term is the present value of a single amount, the final payment of the bond's par value.

$$\text{Bond value} = \text{Coupon} \times \frac{1}{r} \left[1 - \frac{1}{(1+r)^T} \right] + \text{Par value} \times \frac{1}{(1+r)^T}$$

Annuity
factor

Present value
factor

Example 14.2 Bond Pricing ²

What is the value of an 8% semiannual coupon, 30-year maturity bond with par value of \$1,000 if interest rates are 8%?

- Pays 60 semiannual coupon payments of \$40.
- Six-month interest is 4%.
- The value of the bond can be written as:

$$\begin{aligned}\text{Price} &= \sum_{t=1}^{60} \frac{\$40}{(1.04)^t} + \frac{\$1,000}{(1.04)^{60}} \\ &= \$40 \times \text{Annuity factor } (4\%, 60) + \$1,000 \times \text{PV factor } (4\%, 60) \\ &= \$904.94 + \$95.06 = \$1,000\end{aligned}$$

The present value of the bond's 60 semiannual coupon payments of \$40 each is \$904.94.

The \$1,000 final payment of par value has a present value of \$95.06.

Accrued Interest and Quoted Bond Price

- Quoted price does not include interest that accrues between coupon payments.
- Buyer pays seller for accrued interest, the prorated share of the upcoming semiannual coupon:

$$\text{Accrued interest} = \frac{\text{Annual coupon payment}}{2} \times \frac{\text{Days since last coupon payment}}{\text{Days separating coupon payments}}$$

- The sale, or invoice, price of the bond would equal the stated price (flat price) plus the accrued interest:

$$\text{Invoice price} = \text{Flat price} + \text{Accrued interest}$$

Invoice Pricing

What is the invoice price of an 8% semiannual coupon, 30-year maturity bond with par value of \$1,000 if interest rates are 8%? Thirty days have passed since the last coupon payment (assume 182 days in each period).

$$\text{Flat Price} = \sum_{t=1}^{60} \frac{\$40}{(1.04)^t} = \$904.94 + \$95.06 = \$1,000$$

$$\begin{aligned} \text{Accrued interest} &= \frac{\text{Annual coupon payment}}{2} \times \frac{\text{Days since last coupon payment}}{\text{Days separating coupon payments}} \\ &= \frac{\$80}{2} \times \frac{30}{182} = \$6.59 \end{aligned}$$

$$\begin{aligned} \text{Invoice price} &= \text{Flat price} + \text{Accrued interest} \\ &= \$1,000 + \$6.59 = \$1,006.59 \end{aligned}$$

Bond Prices and Yields

- Inverse relationship between price and yield is a central feature of fixed-income securities. (Figure 14.3)
- Interest rate fluctuations represent the main source of risk in the fixed-income market.
- The price curve is convex and becomes flatter at higher interest rates.
- The longer the maturity of the bond, the more sensitive the bond's price to changes in market interest rates.

Figure 14.3 The Inverse Relationship Between Bond Prices and Yields

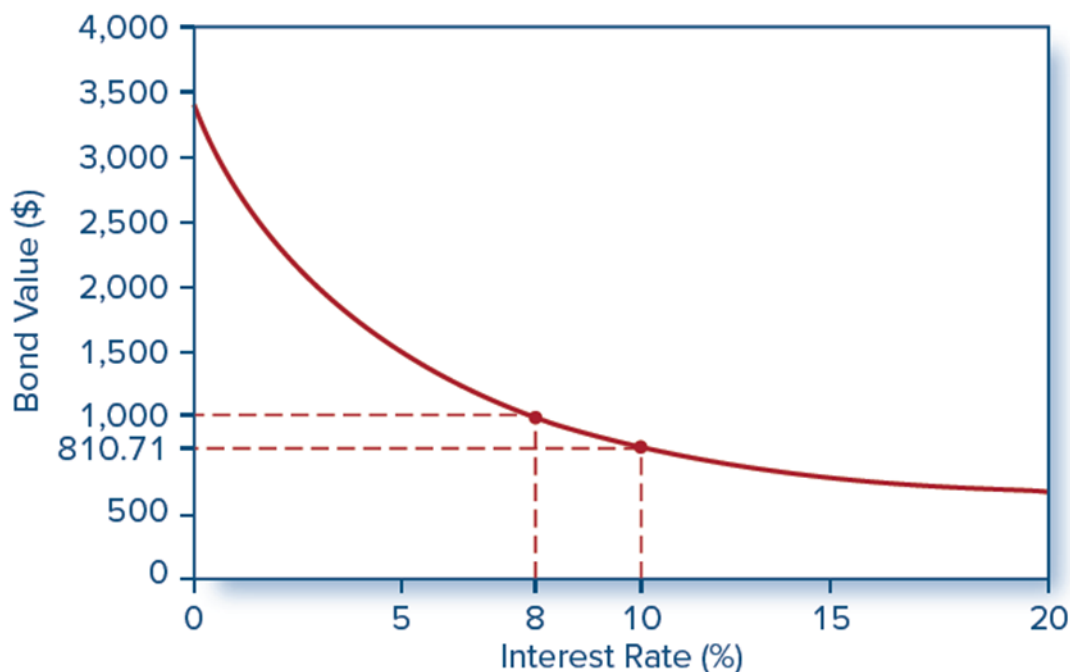


Figure 14.3 The inverse relationship between bond values and the interest rate. Value of an \$8% coupon bond with 30-year maturity making semiannual payments.

Table 14.2 Bond Prices at Different Interest Rates (8% Coupon Semiannual Bond)

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Table 14.2

Bond values at different interest rates (8% coupon bond, coupons paid semiannually)

Time to Maturity	Bond Value at Given Market Interest Rate				
	2%	4%	6%	8%	10%
1 year	1,059.11	1,038.83	1,019.13	1,000.00	981.41
10 years	1,541.37	1,327.03	1,148.77	1,000.00	875.35
20 years	1,985.04	1,547.11	1,231.15	1,000.00	828.41
30 years	2,348.65	1,695.22	1,276.76	1,000.00	810.71

Bond Yields: Yield to Maturity

- **Yield to maturity (YTM)** is the interest rate that makes the present value of a bond's payments equal to its price
- Interpreted as a measure of the average rate of return that will be earned on a bond if it is bought now and held until maturity.
- To calculate YTM, solve the bond price equation for the interest rate given the bond's price.

$$\boxed{\begin{matrix} \text{Market} \\ \text{Bond} \\ \text{Price} \end{matrix}} \text{ Bond Value} = \sum_{t=1}^T \frac{\text{Coupon}}{(1 + \boxed{r_{YTM}})^t} + \frac{\text{Par Value}}{(1 + \boxed{r_{YTM}})^T}$$

Yield to Maturity Example

Suppose an 8% coupon (semiannual), 30-year bond is selling for \$1,150. What is the YTM?

$$\text{Bond Value} = \sum_{t=1}^T \frac{\text{Coupon}}{(1 + r_{YTM})^t} + \frac{\text{Par Value}}{(1 + r_{YTM})^T}$$

$$\$1,150 = \sum_{t=1}^{60} \frac{\$40}{(1 + r_{YTM})^t} + \frac{\$1,000}{(1 + r_{YTM})^{60}} \rightarrow$$

$$r_{YTM} = 6.82\%$$

Bond Yields: YTM Versus Current Yield

- Yield to maturity.
 - Bond's internal rate of return.
 - Compound rate of return over life of the bond assuming all coupons can be reinvested at that yield.
 - Proxy for average return.
- **Current yield** is the bond's annual coupon payment divided by its price.
 - **Premium bonds:** *Coupon rate > Current yield > YTM.*
 - **Discount bonds:** *Coupon rate < Current yield < YTM.*

Bond Yields: Yield to Call

- **Yield to call (YTC)** is calculated like the yield to maturity except that the time until call replaces time until maturity, and the call price replaces par value.
- Low interest rates
 - The price of the callable bond is flat since the risk of repurchase or call is high.
- High interest rates
 - The price of the callable bond converges to that of a normal bond since the risk of call is negligible.

Bond Prices: Callable and Straight Debt

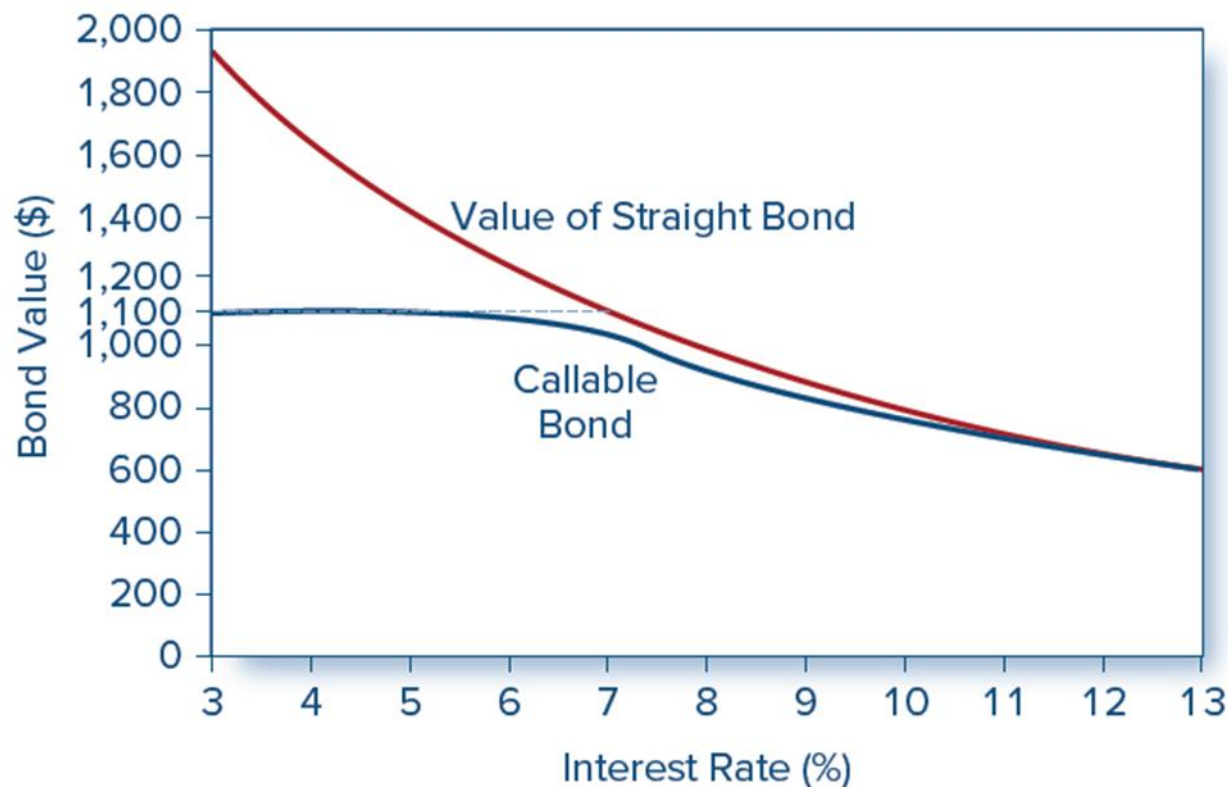


Figure 14.4 Bond values: Callable and straight debt (coupon = \$8; maturity = 30 years; semiannual payments)

Yield to Call Example

	Yield to Call	Yield to Maturity
Coupon payment	\$40	\$40
Number of semiannual periods	20 periods	60 periods
Final payment	\$1,100	\$1,000
Price	\$1,150	\$1,150

Suppose an 8% coupon (semiannual), 30-year bond sells for \$1,150. If the call price is \$1,100 and the time to call is 10 years, what is the yield to call?

$$\$1,150 = \sum_{t=1}^{20} \frac{\$40}{(1+r)^t} + \frac{\$1,100}{(1+r)^{20}} \rightarrow$$

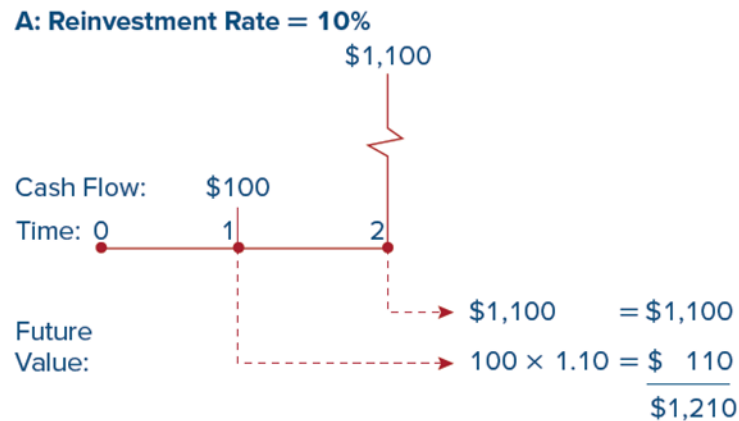
$$r_{YTC} = 6.64\%$$

$$r_{YTM} = 6.82\%$$

Bond Yields: Realized Compound Return versus YTM

- YTM will equal the rate of return realized over the life of the bond if all coupons are reinvested to earn the bond's YTM.
- **Realized compound return** is the compound rate of return assuming that coupon payments are reinvested until maturity.
- Forecasting the realized compound yield over various holding periods or investment horizons is horizon analysis.

Figure 14.5 Growth of Invested Funds

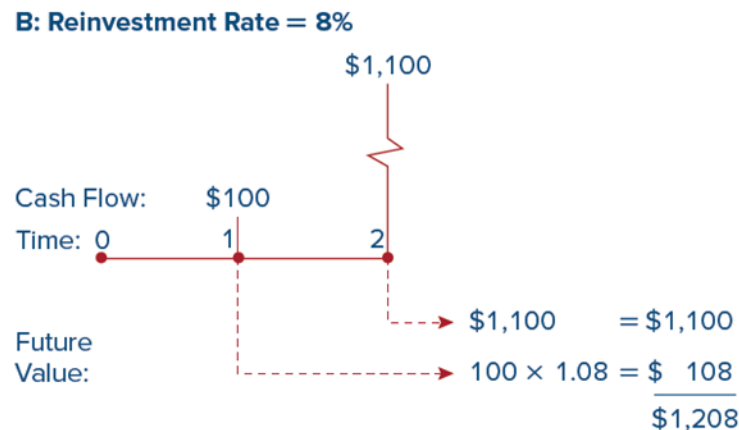


$$V_0(1+r)^2=V_2$$

A:

$$1000(1+r)^2=1210$$

$$r=10\%$$



B:

$$1000(1+r)^2=1208$$

$$r=9.91\%$$

Figure 14.5 Growth of Invested funds.

Horizon Analysis

- Forecasting the realized compound yield over various holding periods or investment horizons is called horizon analysis.
- The forecast of total return depends on your forecasts of both
 - (1) the price of the bond (in turn depends on YTM) when you sell it and at the end of our horizon and
 - (2) the rate at which you are able to reinvest coupon income.

Example 14.9 Horizon Analysis

- Suppose you buy a 30-year, 7.5% (annual payment) coupon bond for \$980 (when its YTM is 7.67%) and plan to hold it for 20 years.
- Your forecast is that the bond's YTM will be 8% when it is sold and that the reinvestment rate on the coupons will be 6%.
- At the end of your investment horizon, the bond will have 10 years remaining until expiration, so the forecast sales price (using a YTM of 8%) will be \$966.45.
- The 20 coupon payments will grow with compound interest to \$2,758.92. (This is the future value of a 20-year \$75 annuity with an interest rate of 6%.)

Example 14.9 Horizon Analysis (cont.)

- Based on these forecasts, your \$980 investment will grow in 20 years to $\$966.45 + \$2,758.92 = \$3,725.37$.
- This corresponds to an annualized compound return of 6.90%:

$$V_0 (1+r)^{20} = V_{20}$$

$$\$980(1+r)^{20} = \$3,725.37$$

$$r = 0.0690 = 6.90\%$$

Figure 14.6 Bond Prices Over Time

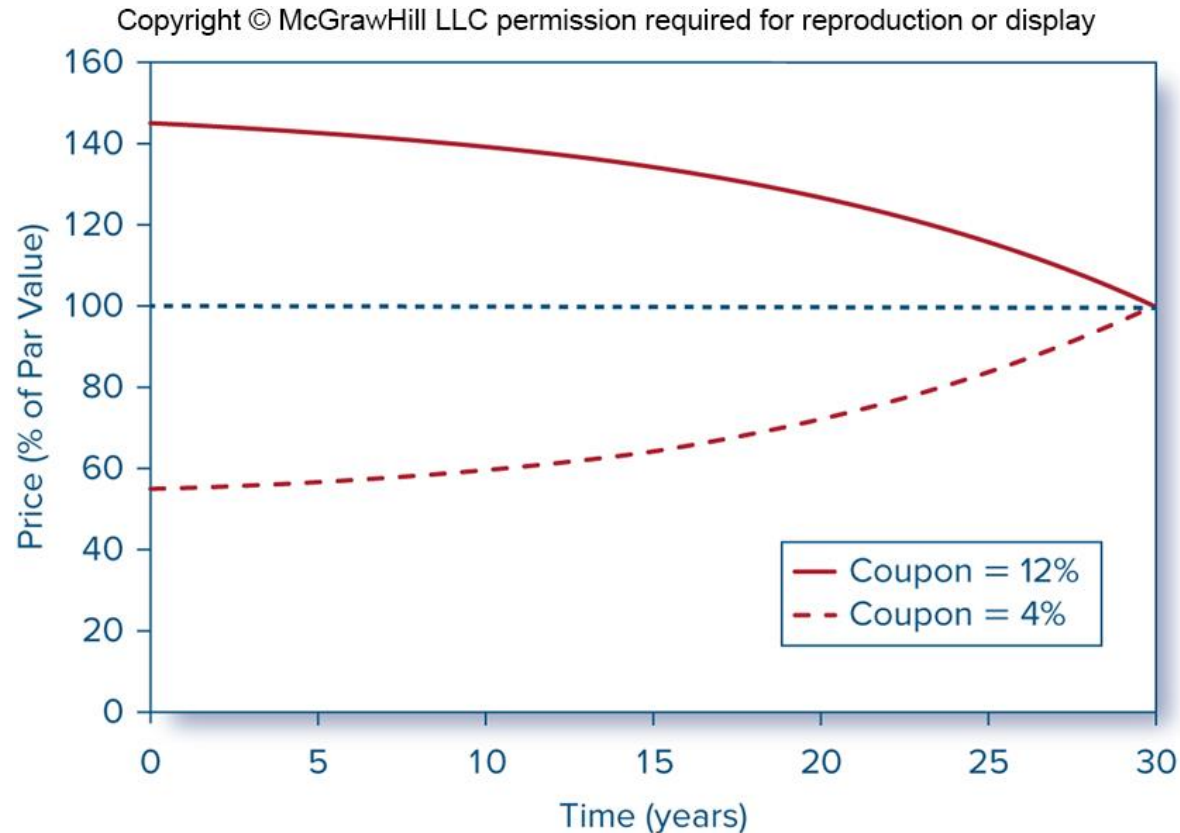


Figure 14.6 Price path of two 30-year maturity bonds, each selling at a yield to maturity of 8%. Bond price approaches par value as maturity date approaches.

Bond Prices Over Time: YTM Versus HPR

YTM

- Average return if the bond is held to maturity.
- Depends on coupon rate, maturity, and par value.
- All of these are readily observable.

$$\text{Bond} = \sum_{t=1}^T \frac{\text{Coupon}}{(1 + r_{YTM})^t} + \frac{\text{Par Value}}{(1 + r_{YTM})^T}$$

HPR

- Rate of return over a particular investment period.
- Depends on the bond's price at the end of the holding period, an unknown future value.

$$HPR = \frac{\text{Sell Price} - \text{Buy Price} + \text{Coupon(s)}}{\text{Buy Price}}$$

Holding-Period Return Example

- CR = 8% YTM = 8% N=10 years
- Semiannual Compounding $P_0 = \$1000$
- In six months the rate falls to 7%

$$\begin{aligned} P_1 &= \text{Annuity factor (3.5\%, 19)} \times 40 \\ &\quad + \text{PV factor (3.5\%, 19)} \times 1000 \\ &= \$1068.55 \end{aligned}$$

$$\text{HPR} = [40 + (1068.55 - 1000)] / 1000$$

$$\text{HPR} = 10.85\% \text{ (semiannual)}$$

OLD bonds

- OLD (original-issue discount) bonds
- Zero-Coupon bond
- Treasury STRIPS (Separated Trading of Registered Interest and Principal of Securities)
- “Built-in” price appreciation as implicit interest payment (Figure 14.7)
- IRS calculates a price appreciation schedule to impute taxable interest income for the built-in appreciation during a tax year.
- If bond is sold, any additional gains or losses that arises from changes in market interest rates are treated as capital gains or losses.

Price of a 30-Year Zero-Coupon Bond Over Time

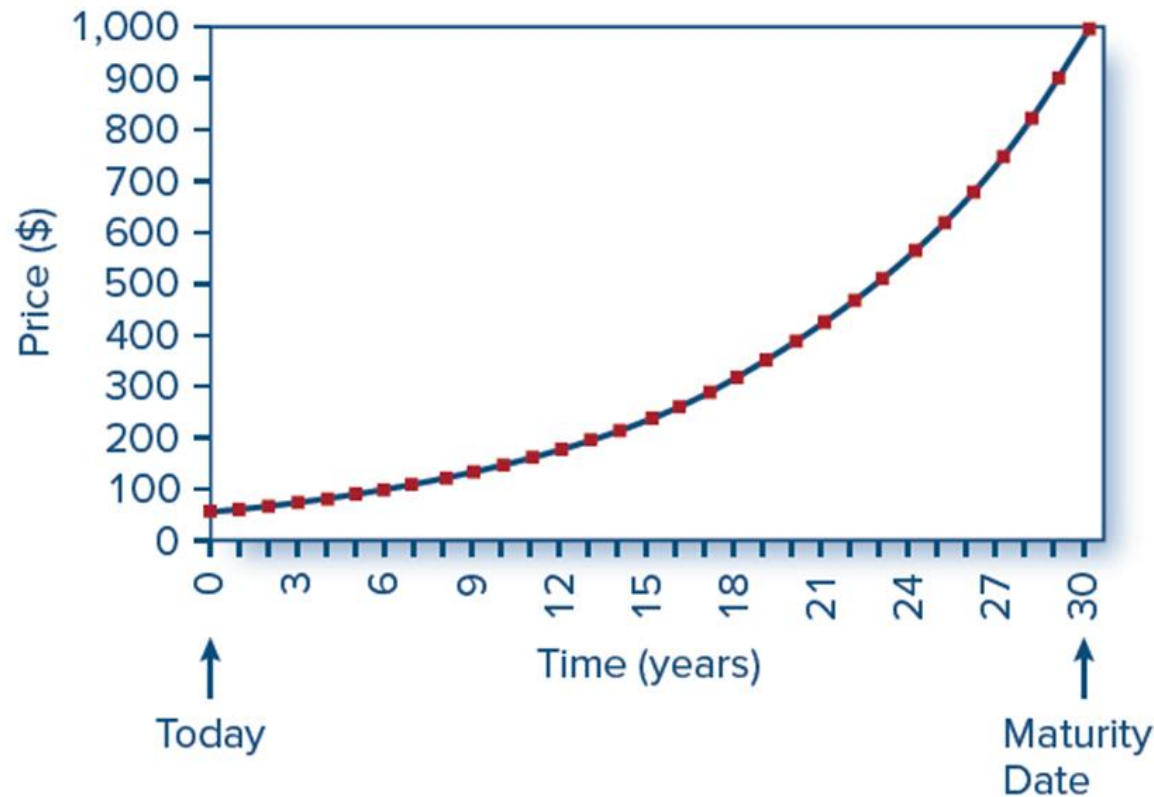


Figure 14.7 The price of a 30-year zero-coupon bond over time at a yield to maturity of 10%. Price equals $\$1,000/(1.10)^T$, where T is time until maturity.

End-of-Chapter exercise - OLD bond example (EOC HW 14.19)

- A newly issued 20-year maturity, zero coupon bond is issued with a yield to maturity of 8% and face value \$1,000. Find the imputed interest income in the first, second, and last year of bond's life.

The price schedule is as follows:

Year	Remaining Maturity (T)	Constant Yield Value $1,000/(1.08)^T$	Imputed interest (Increase in constant yield value)
0 (now)	20 years	\$214.55	
1	19	231.71	$231.71 - 214.55 = 17.16$
2	18	250.25	$250.25 - 231.71 = 18.54$
19	1	925.93	
20	0	1,000	$1,000 - 925.93 = 74.07$

End-of-Chapter exercise - OLD bond example (EOC HW 14.31)

- A newly issued bond pays its coupons once a year. Its coupon rate is 5%, its maturity is 20 years, and its yield to maturity is 8%.
- a. Find the holding-period return for a one-year investment period if the bond is selling at a yield to maturity of 7% by the end of the year.

a. Initial price, $P_0 = 705.46$ [$n = 20$; $PMT = 50$; $FV = 1,000$; $i = 8$]

Next year's price, $P_1 = 793.29$ [$n = 19$; $PMT = 50$; $FV = 1,000$; $i = 7$]

$$\text{HPR} = \frac{\$50 + (\$793.29 - \$705.46)}{\$705.46} = 0.1954 = 19.54\%$$

End-of-Chapter exercise - OLD bond example (EOC HW 14.31) (cont.)

- b. If you sell the bond after one year when its yield is 7%, what taxes will you owe if the tax rate on interest income is 40% and the tax rate on capital gains income is 30%? The bond is subject to original-issue discount (OID) tax treatment.
- c. (10%) What is the after-tax holding-period return on the bond?
- b. Using OID tax rules, the cost basis and imputed interest under the constant yield method are obtained by discounting bond payments at the *original* 8% yield to maturity and simply reducing maturity by one year at a time:

$$P_0 = \$705.46$$

End-of-Chapter exercise - OLD bond example (EOC HW 14.31) (cont.)

First Year $n=19, i=8$

Constant yield price, $P'_1 = \$711.89$, so imputed taxable interest over the first year is: $\$711.89 - \$705.46 = \$6.43$

Coupon received and imputed taxable interest in the year are taxed as the ordinary income: $40\% \times (\$50 + \$6.43) = \$22.57$

Capital gain = Actual price at 7% YTM – Constant yield price = $P_1 - P'_1$
 $= \$793.29 - \$711.89 = \$81.40$
 $n=19, i=7$ $n=19, \underline{i=8}$

Tax on capital gain = $30\% \times \$81.40 = \24.42

Total taxes = $\$22.57 + \$24.42 = \$46.99$

$$\text{c. After-tax HPR} = \frac{\$50 + (\$793.29 - \$705.46) - \$46.99}{\$705.46}$$

$$= 0.1288 = 12.88\%$$

Default Risk and Bond Pricing ¹

- **Credit risk**, or default risk, is the risk the bond will not make all promised payments.
- Rating companies
 - Moody's Investor Service, Standard & Poor's, and Fitch Investor Service.
- Rating categories.
 - Highest rating is AAA (or Aaa).
 - **Investment grade bonds** are rated BBB/Baa or above.
 - **Speculative-grade/junk bonds** are rated below BB B/Baa.

Figure 14.8 Definitions of Each Bond Rating Class

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Bond Ratings									
		Very High Quality		High Quality		Speculative		Very Poor	
Standard & Poor's		AAA	AA	A	BBB	BB	B	CCC	D
Moody's		Aaa	Aa	A	Baa	Ba	B	Caa	C
		At times both Moody's and Standard & Poor's have used adjustments to these ratings: S&P uses plus and minus signs; A+ is the strongest A rating and A- the weakest. Moody's uses a 1, 2, or 3 designation, with 1 indicating the strongest.							
Moody's	S&P								
Aaa	AAA	Debt rated Aaa and AAA has the highest rating. Capacity to pay interest and principal is extremely strong.							
Aa	AA	Debt rated Aa and AA has a very strong capacity to pay interest and repay principal. Together with the highest rating, this group comprises the high-grade bond class.							
A	A	Debt rated A has a strong capacity to pay interest and repay principal, although it is somewhat more susceptible to the adverse effects of changes in circumstances and economic conditions than debt in higher-rated categories.							
Baa	BBB	Debt rated Baa and BBB is regarded as having an adequate capacity to pay interest and repay principal. Whereas it normally exhibits adequate protection parameters, adverse economic conditions or changing circumstances are more likely to lead to a weakened capacity to pay interest and repay principal for debt in this category than in higher-rated categories. These bonds are medium-grade obligations.							
Ba	BB	Debt rated in these categories is regarded, on balance, as predominantly speculative with respect to capacity to pay interest and repay principal in accordance with the terms of the obligation. BB and Ba indicate the lowest degree of speculation, and CC and Ca the highest degree of speculation. Although such debt will likely have some quality and protective characteristics, these are outweighed by large uncertainties or major risk exposures to adverse conditions. Some issues may be in default.							
B	B								
Caa	CCC								
Ca	CC								
C	C	This rating is reserved for income bonds on which no interest is being paid.							
D	D	Debt rated D is in default, and payment of interest and/or repayment of principal is in arrears.							

Finer graduation of ratings

Moody's: 1, 2, 3

Others: + or -

Default Risk and Bond Pricing ²

- Determinants of bond safety.
 - Coverage ratios.
 - Leverage (e.g., debt-to-equity) ratios.
 - Liquidity ratios.
 - Profitability ratios.
 - Cash flow-to-debt ratio.

Table 14.3 Financial Ratios by Rating Class

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	Aaa	Aa	A	Baa	Ba	B	C
EBITA/Assets (%)	12.3%	10.5%	11.2%	9.0%	8.8%	6.6%	4.9%
Operating profit margin (%)	24.8%	18.9%	14.3%	12.9%	11.9%	9.6%	4.6%
EBITA to interest coverage (multiple)	12.0	19.0	12.2	6.5	3.9	2.0	0.8
Debt/EBITDA (multiple)	1.9	1.8	2.1	2.8	3.6	5.1	6.9
Debt/(Debt + Equity) (%)	42.8%	36.6%	39.6%	45.4%	55.7%	65.9%	79.8%
Funds from operations/Total debt (%)	40.4%	44.5%	37.7%	28.2%	21.3%	12.5%	5.7%
Retained Cash Flow / Net Debt (%)	32.3%	29.5%	31.2%	26.6%	21.3%	13.0%	6.0%

Table 14.3

Financial ratios by rating class

Note: EBITA is earnings before interest, taxes, and amortization. EBITDA is earnings before interest, taxes, depreciation, and amortization.

Source: Moody's Financial Metrics, *Key Ratios by Rating and Industry for Global Non-Financial Corporations*, December 2017.

Table 14.3

Financial ratios by rating class

Note: EBITA is earnings before interest, taxes, and amortization. EBITDA is earnings before interest, taxes, depreciation, and amortization.

Source: Moody's Financial Metrics, *Key Ratios by Rating and Industry for Global Non-Financial Corporations*, December 2017.

Figure 14.9 Discriminant Analysis

- Can financial ratios can be used to predict default risk?
- Edward Altman used discriminant analysis to predict bankruptcy.
- Firm is assigned a score based on its financial characteristics.
- If score exceeds cut-off value, the firm is deemed creditworthy.

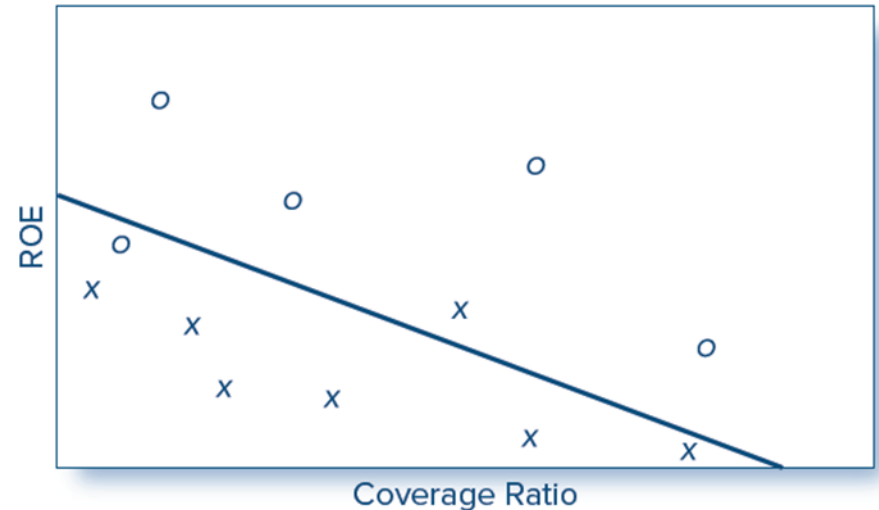


Figure 14.9 Discriminant analysis

Altman Z-Score

$$Z = 3.1 \frac{\text{EBIT}}{\text{Total assets}} + 1.0 \frac{\text{Sales}}{\text{Assets}} + .42 \frac{\text{Shareholders' equity}}{\text{Total liabilities}} \\ + .85 \frac{\text{Retained earnings}}{\text{Total assets}} + .72 \frac{\text{Working capital}}{\text{Total assets}}$$

- Z-scores below 1.23 indicate vulnerability to bankruptcy.
- Scores between 1.23 and 2.90 are a gray area.
- Scores above 2.90 are considered safe.

Bond Indentures

- **Sinking fund** calls for the issuer to periodically repurchase some proportion of the outstanding bonds prior to maturity.
- **Subordination clauses** restrict the amount of additional borrowing by the firm.
- Dividend restrictions limit the payment of dividends by firms.
- **Collateral** is a particular asset that the bondholders receive if the firm defaults.

Figure 14.10 Apple's 2015 Bond Issue

Comment	Description of Bond
1. Interest of 3.45% will be payable on February 9 and August 9 of each year. Thus every 6 months each note will pay interest of $(.0345/2) \times \$1,000 = \17.25 .	→ ISSUE: Apple Inc. 3.45% Notes
2. Investors will be repaid the \$1,000 face value in 2045.	→ DUE: February 9, 2045
3. Moody's bond rating is Aa, the second-highest-quality rating.	→ RATING: Aa
4. A trustee is appointed to look after investors' interest.	→ TRUSTEE: Issued under an indenture between Apple and The Bank of New York Mellon Trust Company
5. The bonds are registered. The registrar keeps a record of who owns the bonds.	→ REGISTERED: Issued in registered, book-entry form
6. The company is not obliged to repay any of the bonds on a regular basis before maturity.	→ SINKING FUND: None
7. The company has the option to buy back the notes. The redemption price is the greater of \$1,000 or a price that is determined by the value of an equivalent Treasury bond.	→ CALLABLE: In whole or in part at any time
8. The notes are senior debt, ranking equally with all Apple's other unsecured senior debt.	→ SENIORITY
9. The notes are not secured; that is, no assets have been set aside to protect the noteholders in the event of default. However, if Apple sets aside assets to protect any other bondholders, the notes will also be secured by these assets. This is termed a <i>negative pledge clause</i> .	→ SECURITY: The notes are unsecured. However, "if Apple shall incur, assume or guarantee any Debt, ... it will secure ... the debt securities then outstanding equally and ratably with ... such Debt."
10. The principal amount of the issue was \$2 billion. The notes were sold at 99.11% of their principal value.	→ OFFERED: \$2,000,000,000 at 99.11%
11. The book runners are the managing underwriters to the issue and maintain the book of securities sold.	→ JOINT BOOK-RUNNING MANAGERS: Goldman, Sachs; Deutsche Bank Securities

YTM and Default Risk

- Must distinguish between the bond's promised YTM and its expected YTM.
- Promised YTM will be realized only if the firm meets the obligations of the bond issue.
- Expected YTM must consider the possibility of a default.
- **Default premium** is a differential in promised yield that compensates the investor for the risk inherent in purchasing a corporate bond that entails some risk of default.

Figure 14.11 Yield Spreads

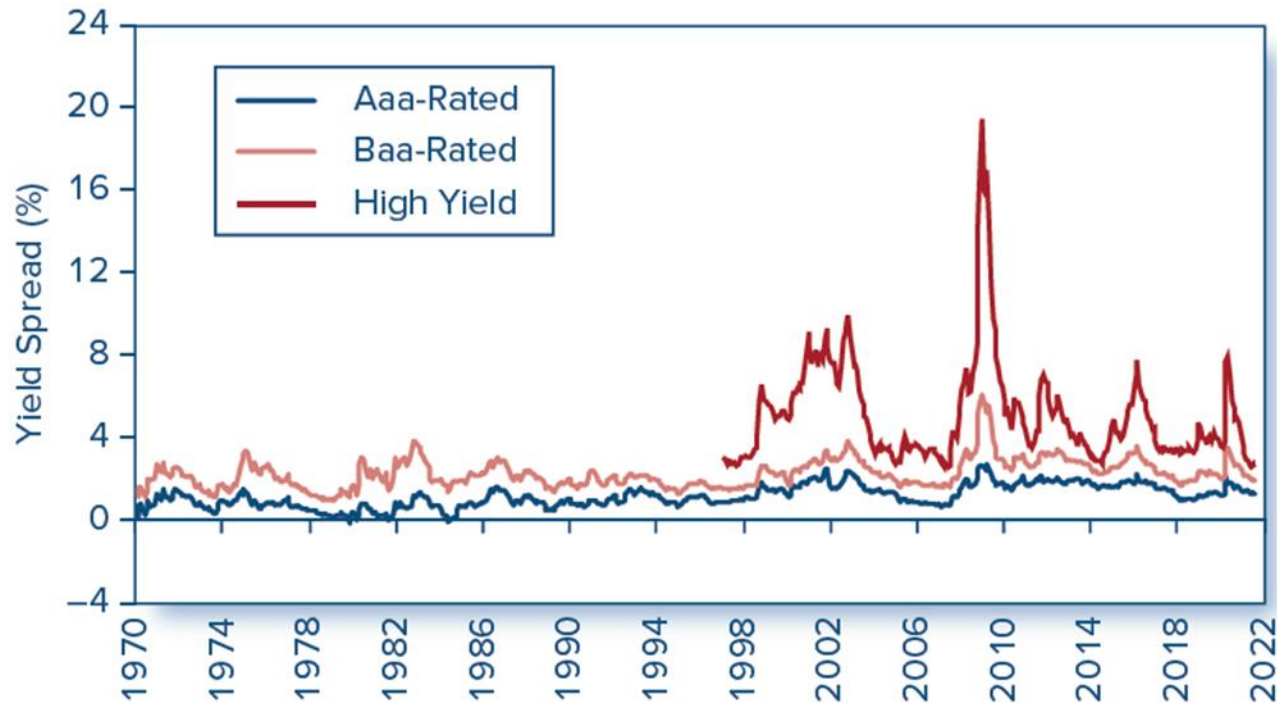


Figure 14.11 Yield spreads between corporate and 10-year Treasury bonds

Source: Federal Reserve Bank of St. Louis.

Example 14.13 Expected vs. Promised YTM

- Suppose a firm issued a 9% coupon bond 20 years ago. The bond now has 10 years left until its maturity date but the firm is having financial difficulties.
- Investors believe that the firm will be able to make good on the remaining interest payments, but that at the maturity date, the firm will be forced into bankruptcy, and bondholders will receive only 70% of par value. The bond is selling at \$750.

Example 14.13 Expected vs. Promised YTM

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	Expected YTM	Stated YTM
Coupon payment	\$45	\$45
Number of semiannual periods	20 periods	20 periods
Final payment	\$700	\$1,000
Price	\$750	\$750

- The YTM based on promised payments is 13.7%. Based on the expected payment of \$700 at maturity, however, the YTM would be only 11.6%.
- The stated YTM is greater than the yield investors actually expect to receive.

Credit Default Swaps (CDSs) ¹

- **Credit default swap (CDS).**
 - Acts like an insurance policy on the default risk of a bond or loan.
 - Allows lenders to buy protection against default risk.
 - Issuer agrees to buy the bond in a default or pay the difference between par and market values to the CDS buyer
 - Risk structure of interest rates and CDS prices ought to be tightly aligned.
 - CDS contracts trade on corporate as well as sovereign debt.

Credit Default Swaps (CDSs) ²

- Natural buyers of CDSs would be large bondholders or banks that wish to enhance the creditworthiness of their outstanding loans.
- But CDSs can also be used to speculate on the financial health of particular issuers:
 - An example of this behavior would be an investor in early 2008 who predicted the imminent financial crisis and purchased CDS contracts on mortgage bonds.

Figure 14.12 Prices of 5-Year Greek Government CDSs



***5-year Greek government bond yield spread over German government bonds**

(German government bond as risk-free benchmark)

Credit Risk and Collateralized Debt Obligations (CDOs)

- **Collateralized debt obligations (CDOs).**
 - Major mechanism to reallocate credit risk in the fixed-income markets.
 - To create a CDO, a legally distinct entity to buy/resell a portfolio of bonds must be established.
 - Structured Investment Vehicle (SIV).
- Loans are pooled together and split into *tranches*.
- Each tranche is given a different level of seniority to claims on the underlying loan pool.
- Mortgage-backed CDOs were an investment disaster in 2007 to 2009.

Figure 14.13 Collateralized Debt Obligations

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		Senior-Subordinated Tranche Structure	Typical Terms
		Senior tranche	70–90% of notional principal, coupon similar to Aa-Aaa rated bonds
		Mezzanine 1	5–15% of principal, investment-grade rating
Bank	Structured investment vehicle, SIV		
		Mezzanine 2	5–15% of principal, higher-quality junk rating
		Equity/first loss/residual tranche	<2%, unrated, coupon rate with 20% credit spread

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